

Sahil Vijendra Jaiswal

DevOps Engineer | SRE | Platform Engineer

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SUMMARY

DevOps and SRE engineer with 2+ years reducing operational overhead at IIT Bombay. Core wins: provisioning automation (6 hours → 12 minutes), deployed GitOps CI/CD shipping 15+ releases weekly, built monitoring detecting 40 incidents before users noticed. Manage 16 Linux servers, containerized microservices on Kubernetes with ArgoCD, handle on-call incident response. Python/Flask backend experience with real-time systems at 800+ msg/sec scale.

TECHNICAL SKILLS

Infrastructure & Automation: Terraform, Ansible, Linux hardening, Nginx, TLS/SSL, firewall rules, DNS, VPN

Containers & Orchestration: Docker, Kubernetes, ArgoCD, GitOps, Helm, multi-node cluster management

CI/CD & DevOps: GitHub Actions, Jenkins, build/test pipelines, automated rollback, drift detection, Git workflows

Observability & SRE: Prometheus, Grafana, alerting, on-call triage, incident response, RCA, runbooks

Cloud & Networking: AWS (EC2, S3, Lambda, IAM, VPC), TCP/IP, load balancing, network security

Backend Development: Python, Bash, Shell, Flask, REST APIs, WebSockets, JWT, MySQL

EXPERIENCE

Project Technical Assistant — DevOps and Platform Engineer

IIT Bombay, Mumbai

Jul 2024 – Present

- **Reduced provisioning time by 98%:** Replaced manual server setup with Terraform and Ansible automation. Environment deployment: 6 hours → 12 minutes. Eliminated configuration drift across 16 Linux servers (RHEL, Ubuntu).
- **40 incidents caught by monitoring before users reported:** Implemented Prometheus metrics and Grafana dashboards with intelligent alerting. Median detection time: 8 minutes. Structured on-call triage reduced MTTR significantly.
- **Scaled deployment cadence from manual to 15+ releases/week:** Designed GitHub Actions CI/CD pipelines for 5 services (build, test, security scan, artifact push). Integrated automated rollback for failed deployments.
- **Architected Kubernetes cluster with GitOps delivery:** Self-managed multi-node Kubernetes cluster. Deployed ArgoCD for declarative continuous delivery. Git as single source of truth. Kubernetes manifests and IaC as reusable Terraform modules.
- **Managed production infrastructure for 1,200+ users:** Own Nginx reverse proxy, TLS lifecycle, load balancing, firewall rules, security hardening, Linux patching across 4 research platforms.
- **Built real-time backend systems at scale:** Developed Python and Flask services (REST APIs, WebSocket channels, JWT auth, RBAC) handling telemetry from 120 Bluetooth devices at 800+ msg/sec. Designed MySQL schemas for persistent storage.
- **Authored operational runbooks and SOPs:** Documented incident playbooks and procedures enabling 4 engineers to independently operate production systems.

PROJECTS

GitOps CI/CD Pipeline on Kubernetes

Kubernetes, ArgoCD, GitHub Actions, Terraform

- Provisioned self-managed multi-node Kubernetes cluster with infrastructure-as-code (Terraform modules for networking, storage, compute).
- Deployed ArgoCD for declarative continuous delivery; Git as single source of truth. GitHub Actions builds Docker images on every commit, runs security scanning, tags with git SHA, pushes to registry. ArgoCD auto-syncs manifests and self-heals on drift.

NeuroScan: Multi-Device Clinical Assessment Platform

Python, Flask, WebSockets, Docker, Kubernetes, MySQL

- Engineered real-time WebSocket server handling telemetry from 120 Bluetooth medical devices at 800 messages per second. Streamed sensor data to concurrent browser dashboards with sub-second latency.
- Implemented MediaPipe pose estimation pipeline for motion capture during clinical assessments. Persistent storage in MySQL. Deployed on Kubernetes with horizontal pod autoscaling based on CPU and custom metrics.
- Containerized entire stack (backend, frontend, database) with Docker Compose for dev/staging and Kubernetes Deployments for production. Monitored with Prometheus custom application metrics.

EDUCATION

M.Sc. Information Technology Guru Nanak Khalsa College (University of Mumbai), Mumbai, CGPA 7.05	2023 – 2025
B.Sc. Information Technology, Distinction Ashmita College (University of Mumbai), Mumbai, CGPA 9.35	2020 – 2023

CERTIFICATIONS

AWS Certified Cloud Practitioner (2024) | Cisco CCNA (2023) | Red Hat Enterprise Linux Administration (2023)